USB Gamepad Standalone Control

1. Course Content

- Use a USB handle connected directly to the robot control board to control the robot's vehicle controller

2. Brief Introduction to the Working Principle

Unlike Section 4.2, which processes handle signals through a chain of ROS2 nodes, the data path in this mode forms a closed loop entirely on the control board:

```
USB handle → control board USB port → firmware parses handle data → directly drives motor/servo/buzzer
```

- After power-up, the firmware automatically detects whether a handle is plugged into the USB port
- Once a handle is detected, it autonomously handles joystick reading, button parsing, and command generation

3. Control Priority and Switching

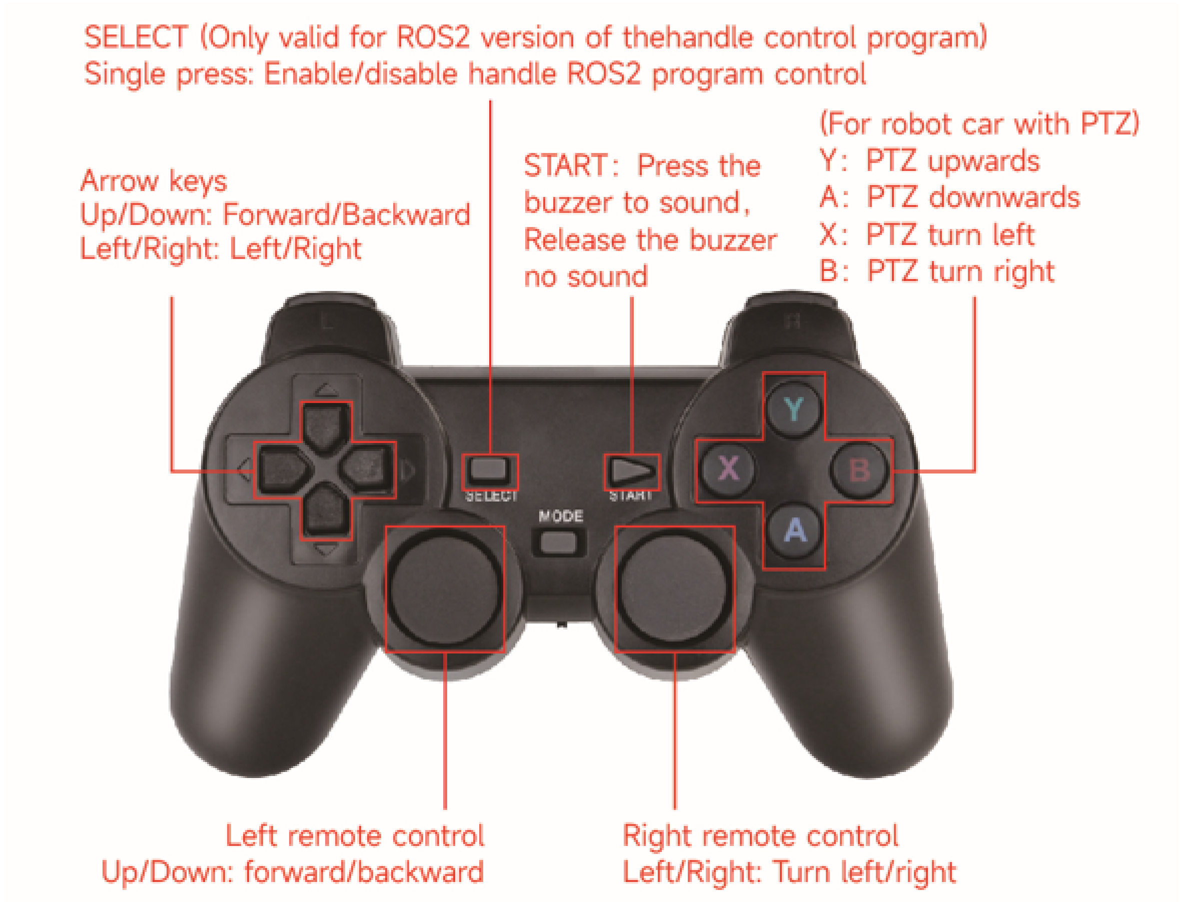
The USB standalone handle and ROS2 topic control can be online at the same time; **the USB standalone handle has higher priority**:

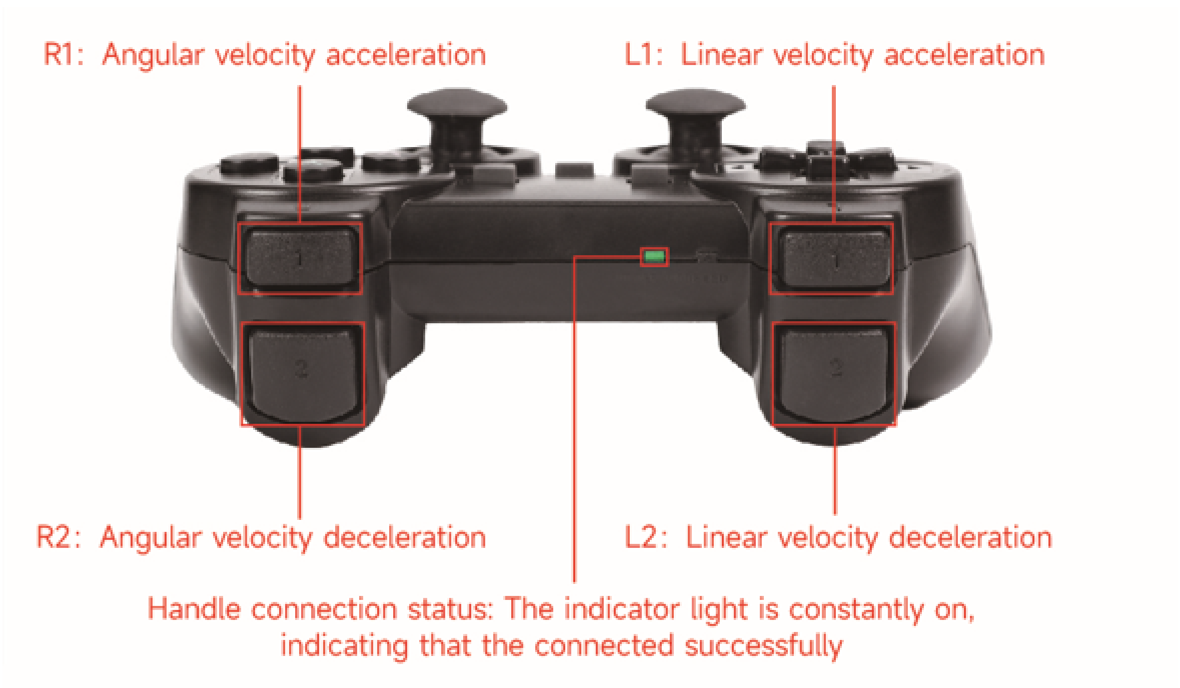
```
Control signal source              Priority   Effective condition
────────────────────────────────────────────────
USB standalone handle (this section)   High   When the handle is plugged into the USB port and has valid input
ROS2 /cmd_vel (4.1/4.2)                 Low    Automatically takes over when there is no USB handle input
```

- When the USB handle has valid joystick input → the firmware uses the USB handle commands and ignores `/cmd_vel`
- When the USB handle has no input (joystick centered or unplugged) → the firmware automatically switches, and `/cmd_vel` takes over

4. Operating Instructions

The button functions are the same as the ROS2 handle control in Section 4.2:





4.1 Movement Control

Input	Function
Left stick up/down	Forward / Backward
Right stick left/right	Turn left / Turn right
D-pad	Digital fallback when the joystick is idle

4.2 Speed Levels

A fixed 5-level speed table (G2 is the default level):

Level	Linear speed (m/s)	Angular speed (rad/s)
G1	0.20	0.50
G2 (default)	**0.30**	**1.50**
G3	0.50	2.50
G4	0.75	3.75
G5	1.00	5.00

Key	Function
L1	Increase linear speed by one level (0→1 rising edge trigger)
L2	Decrease linear speed by one level (0→1 rising edge trigger)
R1	Increase angular speed by one level (0→1 rising edge trigger)
R2	Decrease angular speed by one level (0→1 rising edge trigger)

4.3 Handle Control

Key	Function
X	Servo S1 forward
B	Servo S1 reverse
Y	Servo S2 reverse
A	Servo S2 forward

4.4 Buzzer Feedback

Trigger condition	Behavior
START (OPTIONS) held	Beeps continuously
Gear shift succeeded (up or down one level)	Beeps once briefly
Gear shift hits a boundary (G1/G5)	Beeps twice briefly